

Design inputs

CAR signaling domain

target antigen

binding affinity

checkpoint blockade

TIME scenario

O₂ / lactate / TGF- β

random seed

configure

Application layer

React + Three.js + Chart.js
client-side SPA, ~14k LOC TypeScript

states / config

Simulation layer

spatial ABM (4 cell types) + 10-factor
reaction-diffusion field + checkpoint phagocytosis

context / M1-M2

AI layer

neural surrogate (MLP)
context $\rightarrow$ M1 / M2
 $\approx 8\times$ vs ODE (RK4)



priors / params

Data layer

scRNA-seq + TAM atlas + literature priors
calibratable parameters

observe

Real-time readouts

tumor burden

phagocytosis rate

M1-M2 balance

CD8⁺ activation

spatial cytokine / ECM field

3D cell viewer

Reproducibility & openness

seed 20250706 $\rightarrow$ bit-for-bit runs; benchmark + code ship together